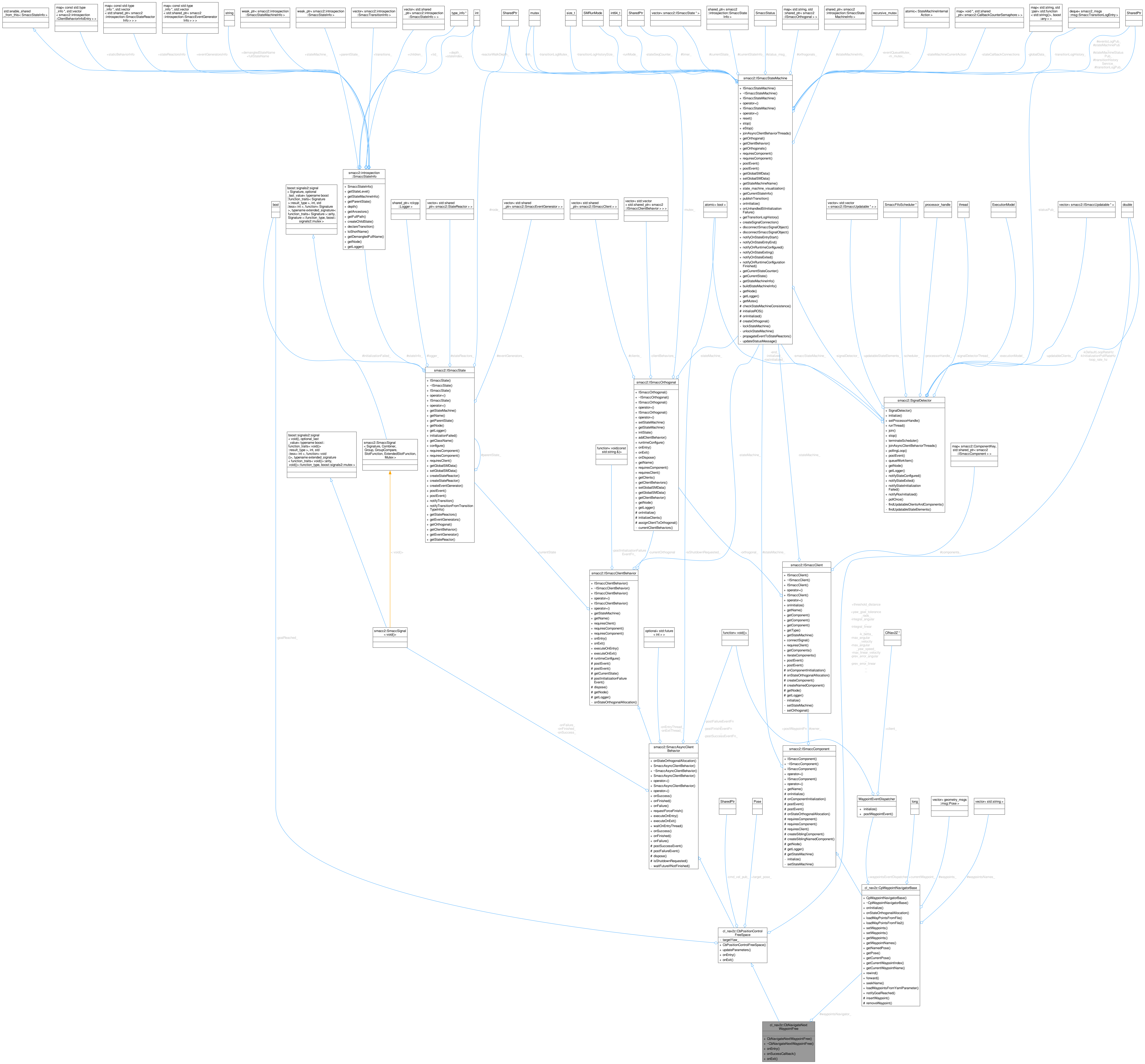
std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
string
weak_ptr< smacc2::introspection::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection::SmaccStateInfo >
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
int
SharedPtr
mutex
size_t
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2::introspection::SmaccStateInfo >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
recursive_mutex
atomic< StateMachineInternalAction >
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
deque< smacc2_msgs::msg::SmaccTransitionLogEntry >
SharedPtr
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+demangledStateName
+fullStateName
+stateMachine_
+parentState_
+transitions_
+children_
+tid_
+depth_
+stateIndex_
-reactorWalkDepth_
#nh_
-transitionLogMutex_
-transitionLogHistorySize_
-runMode_
-stateSeqCounter_
#timer_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
#eventsLogPub_
#stateMachinePub_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ ISmaccStateMachine()
+ operator=()
+ ISmaccStateMachine()
+ operator=()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ onUnhandledStInitializationFailure()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
bool
boost::signals2::signal< Signature, optional_last_value< typename boost::function_traits< Signature >::result_type >, int, std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost::signals2::mutex >
shared_ptr< rclcpp::Logger >
vector< std::shared_ptr< smacc2::StateReactor > >
#node_
vector< std::shared_ptr< smacc2::SmaccEventGenerator > >
vector< std::shared_ptr< smacc2::ISmaccClient > >
vector< std::vector< std::shared_ptr< smacc2::ISmaccClientBehavior > > >
-mutex_
atomic< bool >
vector< std::vector< smacc2::ISmaccUpdatable * > >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
-statusPub_
vector< smacc2::ISmaccUpdatable * >
double
#initializationFailed_
#stateInfo_
#logger_
#stateReactors_
#eventGenerators_
#clients_
-clientBehaviors_
-stateMachine_
-end_
-initialized_
-rosInitialized_
-smaccStateMachine_
-signalDetector_
-updatableStateElements_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-updatableClients_
-kDefaultLoopRateHz
-kInitializationPollRateHz
-loop_rate_hz
smacc2::ISmaccState
+ ISmaccState()
+ ~ISmaccState()
+ ISmaccState()
+ operator=()
+ ISmaccState()
+ operator=()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ initializationFailed()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionTypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
smacc2::ISmaccOrthogonal
+ ISmaccOrthogonal()
+ ~ISmaccOrthogonal()
+ ISmaccOrthogonal()
+ operator=()
+ ISmaccOrthogonal()
+ operator=()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyStateInitializationFailed()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
boost::signals2::signal< void(), optional_last_value< typename boost::function_traits< void()>::result_type >, int, std::less< int >, function< void()>, typename extended_signature< function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
function< void(const std::string &)>
-stateMachine_
-stateMachine_
< void()>
-currentState
-postInitializationFailureEventFn_
-currentOrthogonal
-isShutdownRequested_
-orthogonal_
#stateMachine_
#components_
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ ISmaccClientBehavior()
+ operator=()
+ ISmaccClientBehavior()
+ operator=()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# postInitializationFailureEvent()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ ISmaccClient()
+ operator=()
+ ISmaccClient()
+ operator=()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
+threshold_distance_
+yaw_goal_tolerance_rads_
-integral_angular_
-integral_linear_
-k_betta_
-max_angular_velocity
-max_angular_yaw_speed_
-max_linear_velocity
-prev_error_angular_
-prev_error_linear_
...
smacc2::SmaccSignal< void()>
-goalReached_
optional< std::future< int > >
function< void()>
ClNav2Z *
-onFailure_
-onFinished_
-onSuccess_
-onEntryThread_
-onExitThread_
-postFailureEventFn_
-postFinishEventFn_
-postSuccessEventFn_
+postWaypointFn
#owner_
+client_
smacc2::SmaccAsyncClientBehavior
+ onStateOrthogonalAllocation()
+ SmaccAsyncClientBehavior()
+ ~SmaccAsyncClientBehavior()
+ SmaccAsyncClientBehavior()
+ operator=()
+ SmaccAsyncClientBehavior()
+ operator=()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ ISmaccComponent()
+ operator=()
+ ISmaccComponent()
+ operator=()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
WaypointEventDispatcher
+ initialize()
+ postWaypointEvent()
SharedPtr
Pose
long
vector< geometry_msgs::msg::Pose >
vector< std::string >
-cmd_vel_pub_
+target_pose_
+waypointsEventDispatcher
+currentWaypoint_
#waypoints_
#waypointsNames_
cl_nav2z::CpWaypointNavigatorBase
+ CpWaypointNavigatorBase()
+ ~CpWaypointNavigatorBase()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ loadWayPointsFromFile()
+ loadWayPointsFromFile2()
+ setWaypoints()
+ setWaypoints()
+ getWaypoints()
+ getWaypointNames()
+ getNamedPose()
+ getPose()
+ getCurrentPose()
+ getCurrentWaypointIndex()
+ getCurrentWaypointName()
+ rewind()
+ forward()
+ seekName()
+ loadWaypointsFromYamlParameter()
+ notifyGoalReached()
# insertWaypoint()
# removeWaypoint()
cl_nav2z::CbPositionControlFreeSpace
- targetYaw_
+ CbPositionControlFreeSpace()
+ updateParameters()
+ onEntry()
+ onExit()
#waypointsNavigator_
cl_nav2z::CbNavigateNextWaypointFree
+ CbNavigateNextWaypointFree()
+ ~CbNavigateNextWaypointFree()
+ onEntry()
+ onSucessCallback()
+ onExit()